

Embrace Edge AI, Empower Industrial Digitalization



EC3320 Series AI Edge Computer

· Strong Computing

· High Security

· High Reliability

· Cloud-Managed

1. Product Overview

EC3320 is a high-performance AI edge computer built for industrial local computing, real-time decision-making, and secure cloud-connected operations.

Key features:

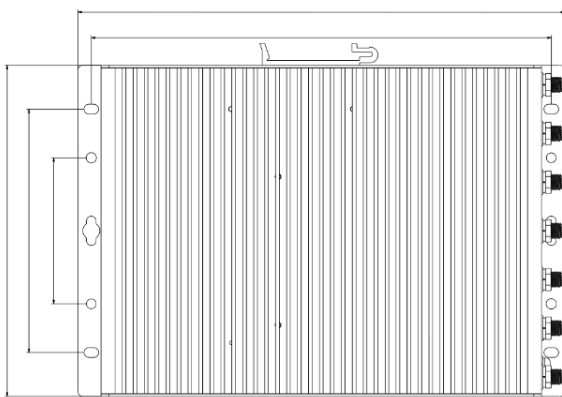
- **High computing capability:** RK3588-based platform with multi-core CPU, high-performance GPU, and up to 6 TOPS NPU
- **Scalable AI acceleration:** Supports Hailo-8 module expansion up to 26 TOPS
- **Rich industrial interfaces:** 3×GE, serial, CAN, DI/DO, HDMI, USB3.0, audio I/O
- **Secure and reliable:** Secure Boot, TPM2.0 (Optional), watchdog, dual-SIM failover
- **Cloud-managed operations:** DeviceLive remote monitoring and edge app/container management

Core Technical Specifications

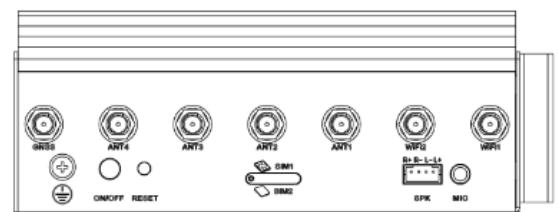
Technical Indicator	Specification
Open Platform	Linux with edge app/container management
CPU	4×Cortex-A76 + 4×Cortex-A55, up to 2.4GHz
GPU	Mali-G610 MC4
NPU	Up to 6 TOPS (INT8), Hailo expansion up to 26 TOPS
Memory / Storage	8GB / 64GB eMMC
Data Acquisition	Modbus RTU/TCP, EtherNet/IP, OPC UA, DNP3.0, BACnet, CNC
Cellular Network	5G NR / 4G Cat.6 (model-dependent)
Network Features	Dual-SIM backup, multi-level link detection, auto-redial

Technical Indicator	Specification
Security (TPM2.0 Optional)	Secure Boot, TPM2.0
Cloud Management	DeviceLive, HTTP/HTTPS/SSH remote management
Interfaces	3×GE, 2×RS-232 + 2×RS-485, CAN, DI/DO, USB3.0, HDMI, MIC/SPK
Power Input	9~36V DC

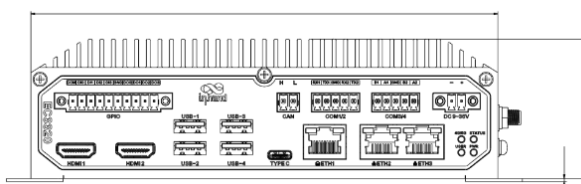
2. Product Dimensions



Front View



Side View



Interface Diagram

Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and **for reference only**.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Platform	
CPU	Quad Cortex-A76 + Quad Cortex-A55, up to 2.4GHz
GPU	Quad-core Mali-G610 MC4 high-performance GPU
NPU	Up to 6 TOPS (INT8) Support INT4/INT8/INT16 mixed operations and framework switching of TensorFlow / MXNet / PyTorch / Caffe
RAM	8GB
FLASH	64GB eMMC
Connectivity & Interfaces	
Ethernet Ports	3×10/100/1000 Mbps GE
I/O Ports	10pin industrial green terminal (4IN, 4OUT, 2GND)
Serial Ports	2×RS-232 + 2×RS-485, industrial terminal
CAN	1×CAN2.0
Buttons	1×Power, 1×Reset
SIM Card Holders	2×Nano-SIM
LED Indicators	Power, Status, 4G, USER
USB	4×USB3.0

Category/Parameter	Specification
Expansion Interfaces	1x M.2(Supports SATA3.0 SSD, supports 2242 size) 1x M.2(Supports Hailo-8 M.2 module, up to 26TOPs computing power)
HDMI	2×HDMI2.0
MIC	Supported
SPK	Supported
Wi-Fi	STA, 802.11ac/a/b/g/n, 2.4G/5G dual band
Bluetooth	BLE5.0
GPS	Supported (requires cellular module support)
Power & Power Consumption	
Input Voltage	9~36V DC (2pin green industrial terminal, with flange)
Power Interface	Industrial DC terminal input
Mechanical Specifications	
Product Dimensions	180 × 136 × 54 mm (7.09 × 5.35 × 2.13 in)
Mounting Method	DIN-rail / wall mounting
Protection Rating	IP40
Enclosure & Heat Dissipation	Metal enclosure, fanless design
TPM (Optional)	TPM2.0
Environment & Certifications	

Category/Parameter	Specification
Storage Temperature	-40 °C ~ 85 °C (-40 °F ~ 185 °F)
Operating Temperature	-20 °C ~ 60 °C (-4 °F ~ 140 °F)
Environmental Humidity	5~95% RH (non-condensing)
Physical Characteristics	IEC60068-2-27 shock resistance IEC60068-2-6 vibration resistance IEC60068-2-32 drop resistance
EMC Standard	EN61000-4-2, level 3, Static EN61000-4-3, level 3, Radiation Electric Field EN61000-4-4, level 3, Pulsed Electric Field EN61000-4-5, level 3, Surge EN61000-4-6, level 3, Conducted Disturbance Immunity EN61000-4-8, Power Frequency Field Resistance, horizontal / vertical 400A/m (>level 2) EN61000-4-12, level 3, Shock Wave Resistance

4. Software Specifications

Category/Parameter	Specification
Operating System	
Operating System	Linux
Network Features	
Network Access	5G NR / LTE Cat.6 (model-dependent), Ethernet
Security	
Secure Boot	Supported

Category/Parameter	Specification
Reliability	
Link Detection	Multi-level link detection with auto-redial
Built-in Watchdog	Embedded watchdog
Dual-SIM Switchover	Supported
Data Acquisition Protocols (DSA)	
Industrial Protocols	Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINSUDP, HostLink, PPI
Electricity Protocols	DLT645-2007, IEC101/104, DNP3.0
Other Protocols	BACnet, CNC
Network Management	
Upgrade Method	Supports patent upgrade mechanism, local or remote firmware upgrade
Log Functions	Local/remote logs
Remote Management	DeviceLive / HTTP / HTTPS / SSH remote management
DeviceLive Cloud	Supports cloud-based parameter configuration, container management, application and firmware management

5. Ordering Information

Model Rule

Model code: EC3320-<WMNN>-[XY]

<WMNN>: Cellular Type & Frequency Band

[XY] (Optional): AI Expansion Module

Model List

Model	Region	Cellular Bands	Ethernet/Serial	GPS
EC3320-NRQ3	Global	5G NR SA/NSA: n1/n2/n3/n5/n7/n8/n12/n20/n 25/n28/n38/n40/n41/n48/n6 6/n71/n77/n78/n79 LTE FDD: B1/B2/B3/B5/B7/B8/B12(B17) /B13/B14/B18/B19/B20/B25/B 26/B28/B29/B30/B32/B66/B 71 LTE TDD: B34/B38/B39/B40/B41/B42/ B48 LAA: B46 WCDMA: B1/B2/B3/B4/B5/B6/B8/B19	3×1000Mbps; 2×RS-232 + 2×RS-485	YES

EC3320-FQ09	Global	4G Cat.6 LTE FDD: B1/B2/B3/B4/B5/B7/B8/B12/ B13/B14/B17/B18/B19/B20/B2 5/B26/B28/B29/B30/B32/B6 6/B71 LTE TDD: B34/B38/B39/B40/B41/B42/ B43/B46(LAA)/B48(CBRS)	3×1000Mbps; 2×RS-232 + 2×RS-485	YES
EC3320-EN00	No Cellular	No Cellular	3×1000Mbps; 2×RS-232 + 2×RS-485	NO

Memory/Storage: 8GB/64GB

Note: All models support Wi-Fi/BT/CAN/MIC/SPK/IO.

AI Expansion Module (Optional)

[XY] P/N Code	Feature
—	None
H8	Hailo-8, M.2 Key B+M 2280

6. Contact Us

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